

WORKED EXAMPLE

A line makes a numerical prediction

Our goal: measure how well one chosen line fits three observed points.

Data: $(x, y) = (1, 2), (2, 3), (3, 5)$. x is an input; y is its known target.

$$h_{\theta}(x) = \theta_0 + \theta_1 x \quad (\text{Lecture 1, p. 33})$$

h is the prediction. θ names the model parameters: θ_0 is the intercept (bias); θ_1 is the slope (weight).

$$\theta_0 = 0, \quad \theta_1 = 1, \quad m = 3 \quad \implies \quad h_{\theta}(x) = x$$

m counts examples. One shared line predicts all three points. These starting parameters are chosen for this study example.

WORKED EXAMPLE

Predict all three targets

Substitute each input into the same line. Do not change the parameters.

$$h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$

The superscript (i) identifies an example; it is not an exponent.

$$h_{\theta}(x^{(1)}) = 0 + 1 \times 1 = 1 \quad y^{(1)} = 2$$

$$h_{\theta}(x^{(2)}) = 0 + 1 \times 2 = 2 \quad y^{(2)} = 3$$

$$h_{\theta}(x^{(3)}) = 0 + 1 \times 3 = 3 \quad y^{(3)} = 5$$

Predictions (1, 2, 3) differ from targets (2, 3, 5). Now measure the gaps.

WORKED EXAMPLE

Keep the sign, then square

Residual means prediction minus target. A negative residual is an underprediction.

$$r_i = h_{\theta}(x^{(i)}) - y^{(i)}$$

$$r_1 = 1 - 2 = -1 \quad r_1^2 = (-1)^2 = 1$$

$$r_2 = 2 - 3 = -1 \quad r_2^2 = (-1)^2 = 1$$

$$r_3 = 3 - 5 = -2 \quad r_3^2 = (-2)^2 = 4$$

Squaring makes every contribution nonnegative. A gap twice as large contributes four times as much squared error.

Keep the signed residuals too: the next lesson needs them for gradients.

WORKED EXAMPLE

Combine errors into one cost

The lecture uses half the mean squared error: be careful with the factor 2.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Lecture 1, PDF page 36. The summation sign means add all example contributions.

$$J = \frac{1}{2 \times 3} (1 + 1 + 4) = \frac{6}{6} = 1$$

$$\text{MSE} = \frac{1+1+4}{3} = 2 \quad J = \frac{1}{2} \text{MSE} = 1$$

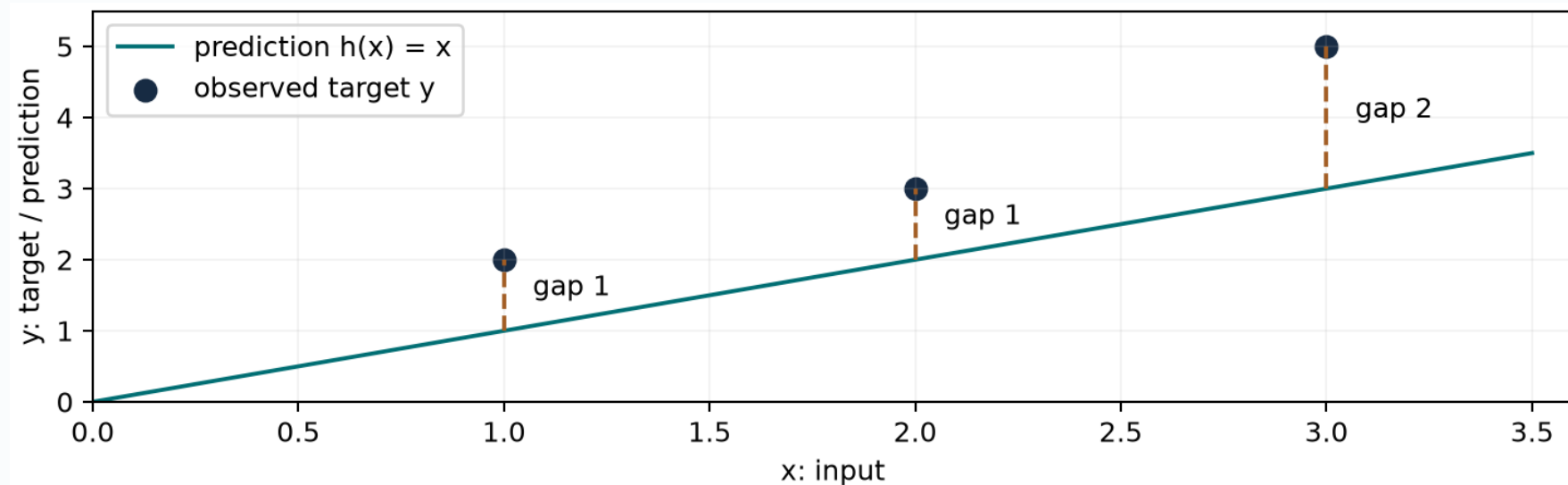
The factor 1/2 keeps the best-fit line the same and simplifies derivatives later. It is a convention, not part of every definition of MSE.

A smaller J means a closer fit under this squared-error measure.

WORKED EXAMPLE

Picture what the cost measures

The vertical gaps are target minus prediction; our signed residual is the reverse.



All three predictions lie below their targets. Squaring the gaps loses their direction; cost alone does not tell us how to move the line.

INDEPENDENT PRACTICE

Your turn: shift the line upward

Stop here and calculate before looking at the answer page.

Use the same three pairs: (1, 2), (2, 3), (3, 5). Change only the intercept.

$$\theta_0 = 1, \quad \theta_1 = 1, \quad h_{\theta}(x) = 1 + x$$

1. Calculate all three predictions.
2. Calculate each residual: prediction minus target.
3. Square the residuals and add them.
4. Divide by 2m. Is this cost lower than the original $J = 1$?

This is a new candidate line, not a gradient-descent update yet.

WORKED ANSWER

Practice answer and common mistakes

Compare each intermediate result with your working.

$$h_1 = 1 + 1(1) = 2, \quad h_2 = 1 + 1(2) = 3, \quad h_3 = 1 + 1(3) = 4$$

$$r_1 = 2 - 2 = 0, \quad r_2 = 3 - 3 = 0, \quad r_3 = 4 - 5 = -1$$

$$J = \frac{0^2 + 0^2 + (-1)^2}{2 \times 3} = \frac{1}{6} \approx 0.166667 < 1$$

This candidate fits better overall. It predicts the first two targets exactly and underpredicts the third by 1.

Common mistakes: dividing by 3 instead of 6; treating $(-1)^2$ as -1 ; using a target as the model prediction.

Next: calculate gradients to choose a useful parameter change automatically.